



Claim-Aligned Assurance for Trustworthy AI : Auditable Privacy and Secure Public Verification

Ph.D. Proposal

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International Publications

1st-author

- PP-Mark: Provable and Publicly Verifiable Watermarking for Generative AI
(NeurIPS 2026 Main Track, 정보과학회(BK21) 최우수 국제학회(Awaiting Final Decision, 5/5/3))
- Ethereum Smart Contract Account Classification and Transaction Prediction Using the Graph Attention Network
(Journal of Web Engineering(SCI) 2024, Q3, IF 1.6)
- Space Authentication in the Metaverse (IEEE Access(SCIE) 2024, Q2, IF 4.2)

Co-author

- LLM-Based Persona-Driven Text Data Augmentation (IEEE Access(SCIE) 2025, Q2, IF 4.2)
- AI detection logic to prevent drug purchase (SCOPUS, New Phys.: Sae Mulli 2025)

Domestic Publications

1st-author

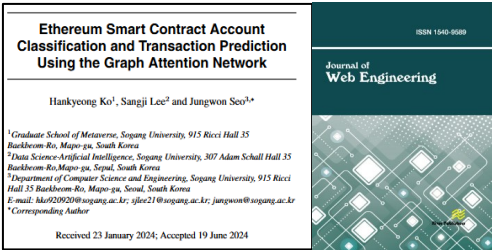
- 메타버스 환경에서의 유스케이스 기반 비기능적 요구사항 테스트 가이드 (2024 Korea Society of Blockchain)
- P2P 전력 거래를 위한 전기차 충전 인프라 구현 방안 (2024 Korea Society of Blockchain)
- 메타버스 환경에서의 공간 기반 SBT 접근 제어 기법 (2023 Korea Society of Blockchain)



Research Trajectory toward Claim-Aligned Assurance

Hidden System Structure

- ANALYZING HIDDEN STRUCTURE
- Ethereum heterogeneous graphs



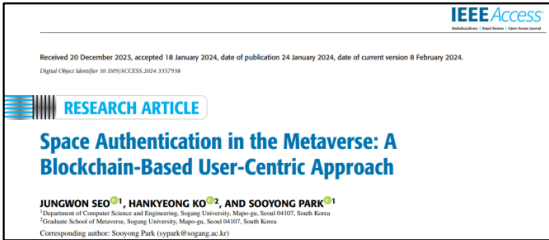
Anonymous transactions → roles and relations

INFER



User-Centric Verification

- REDUCING CENTRALIZED TRUST
- User-centric metaverse authentication



User-managed token + smart-contract verification

VERIFY



Context-Preserving AI Generation

- PRESERVING DOMAIN CONTEXT
- Persona-driven LLM augmentation



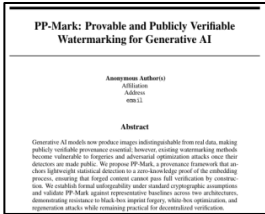
Personas + behavioral rules
→ context-coherent generation

PRESERVE



Claim-Aligned Assurance

- CLAIM-ALIGNED ASSURANCE



Trustworthiness claims aligned with
Object · Mechanism · Interface · Evidence

ALIGN

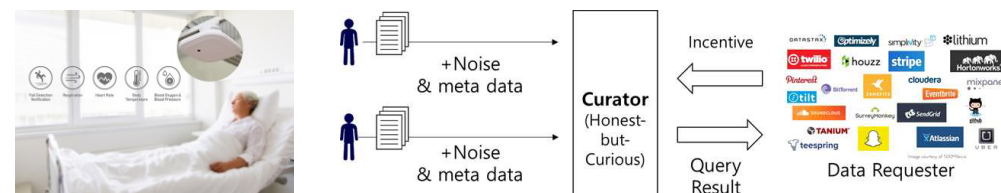
Why Trustworthy AI Claims Fail

Reported Claim $\neq$ Actual Guarantee

- | | |
|---|---|
| <ul style="list-style-type: none"> • User-level Privacy • Authentic AI-generated content • Verified provenance | <ul style="list-style-type: none"> • Protected object • Executed mechanism • Public interface • Verification evidence |
|---|---|

• Privacy

In differential privacy, an (ϵ, δ) pair accounted for generated windows may be reported as patient- or user-level protection.



• Watermark Provenance

A positive watermark score may not prove a legitimate generation process.

내년 1월부터 AI 생성물에 '눈에 보이는' 워터마크 의무화

문지혜 기자 2025.11.12 12:00

구글에서 AI-생성물 추가하기

과학기술정보통신부, AI 기본법 시행령 입법예고
"비가시적 워터마크 넣는 경우에도 AI 생성물 고지해야"

DEEPFAKE

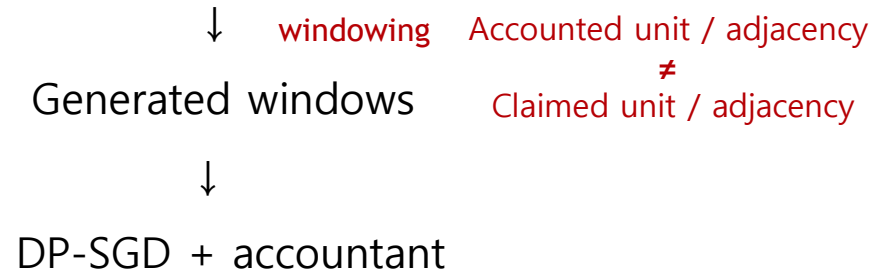


Two Forms of Claim-Guarantee Misalignment

Auditable Privacy

- In Differential Privacy
- "Patient or user level (ϵ , δ) DP"

Window / Raw events / patients / users



Secure Public Verification

- "This content has verified AI provenance."

Public Detector	vs	Private Verification API
Auditability ✓		Detector Exposure ↓
Attack surface ↑		Independent Public Verification ✗

Same structural problem : Claim without mechanism and evidence bound justification

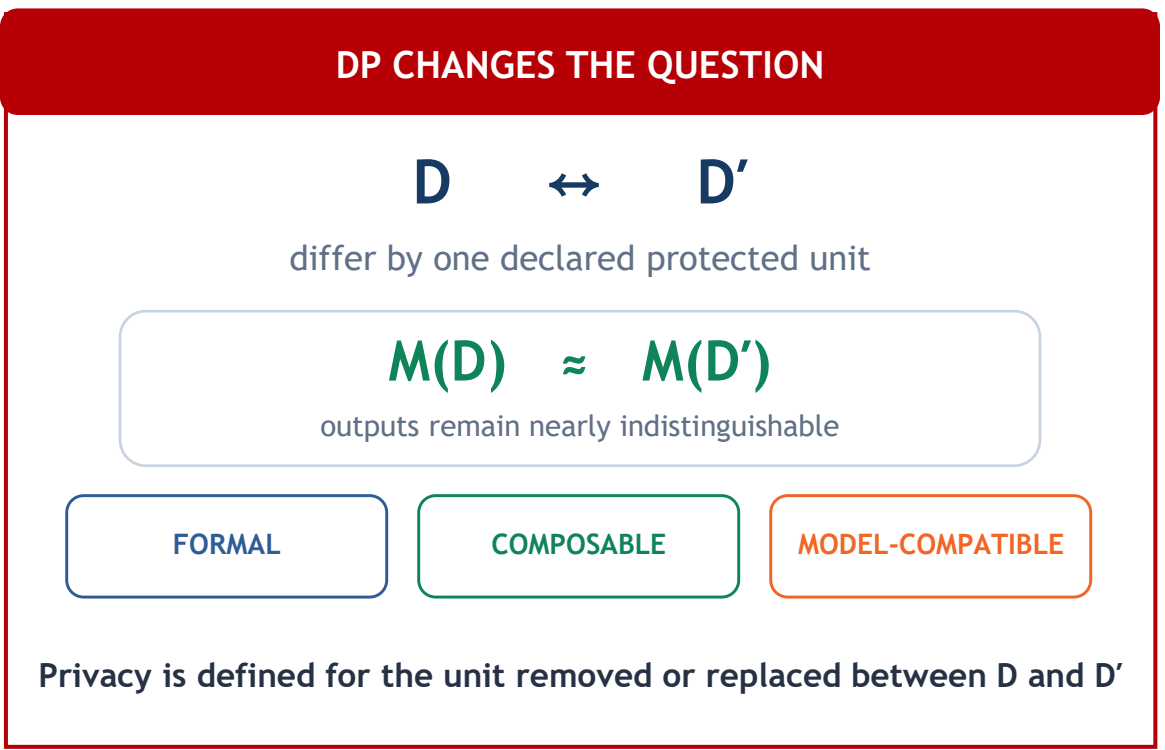
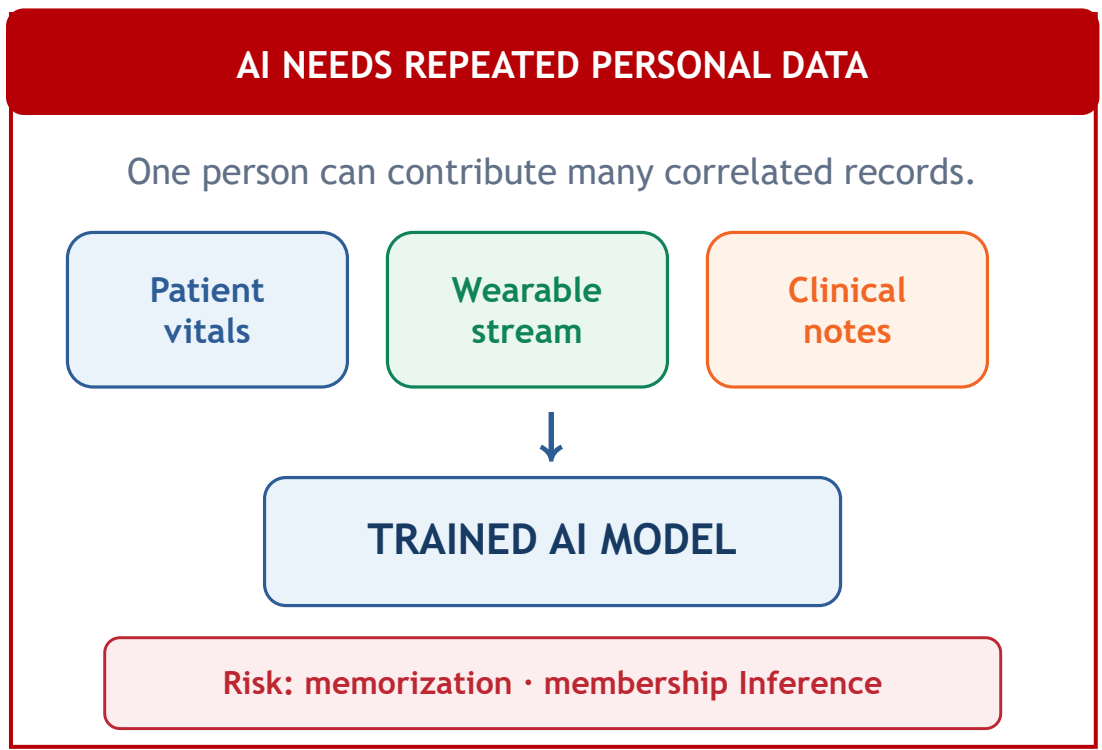


Dissertation Roadmap: Core Studies and Future Validation

RESEARCH AXIS	CURRENT CORE STUDIES	FUTURE VALIDATION
AUDITABLE PRIVACY	Chapter - Claim Validation Determine which privacy-unit claims are supported by evidence from a completed DP-SGD execution.	Future Study A: - Compare post-hoc privacy-unit conversion with native patient-level DP on the same medical task. - When is each claim valid, and at what utility cost?
SECURE PUBLIC VERIFICATION	Chapter - PP-Mark (Publicly Verifiable Mark) Combine robust statistical screening with proof-backed verification, rather than relying on a detector score alone.	Future Study B: Store the proof at publication, retrieve the correct proof when verification is requested.



Why Differential Privacy Entered AI



Medical sequence(ex.): 1 patient → thousands of events → many overlapping windows
What is exactly protected?



Differential Privacy: What Do (ϵ, δ) Mean?

APPROXIMATE DIFFERENTIAL PRIVACY

$$\Pr[M(D) \in S] \leq e^\epsilon \cdot \Pr[M(D') \in S] + \delta$$

for every measurable output set S and every declared neighboring pair D, D'

Intuition: observing the model should not reveal much about the presence of one protected unit.

ϵ • PRIVACY LOSS

Lower $\epsilon \rightarrow$ stronger indistinguishability
Not a privacy percentage

δ • ADDITIVE SLACK

A small failure allowance
Must be meaningful for the protected population

THE NUMBER IS INCOMPLETE WITHOUT UNIT + ADJACENCY

WINDOW

one training example

RAW EVENT

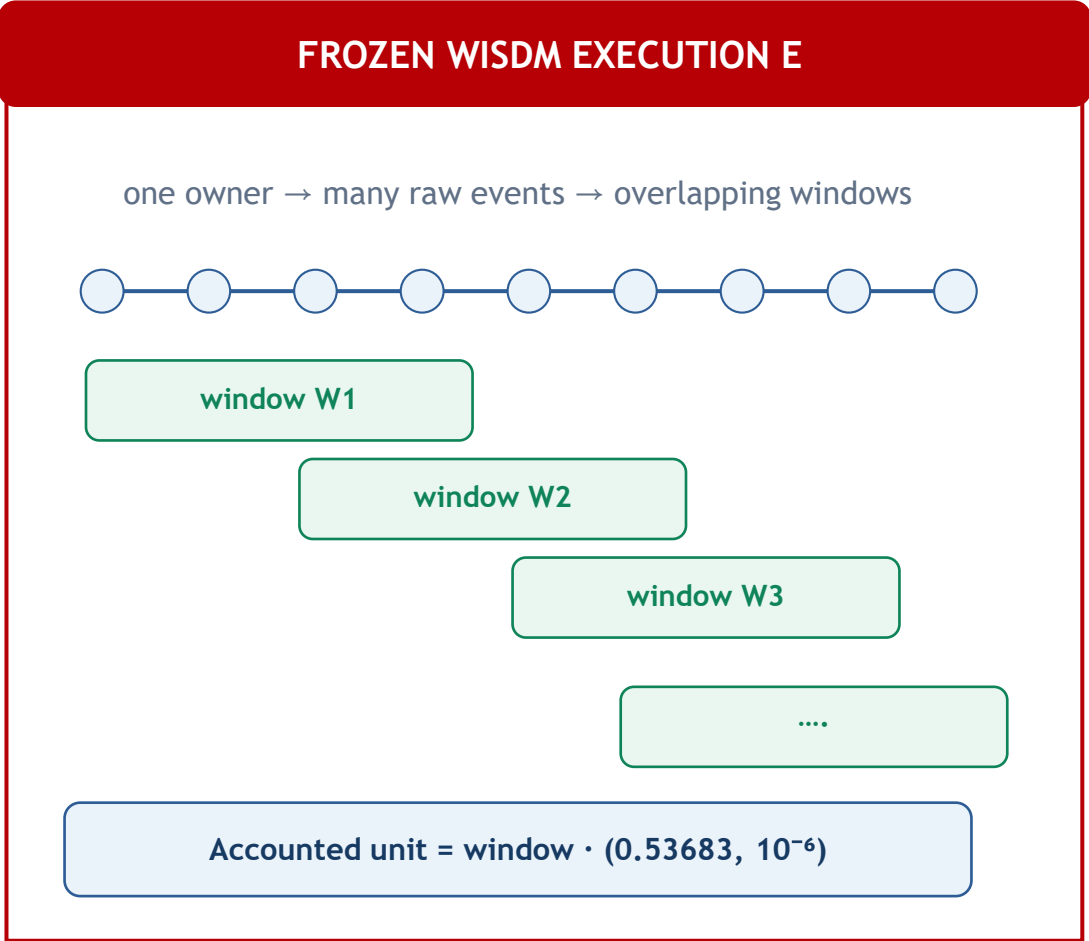
one sensor observation

PATIENT / OWNER

all records from one person



One Run, Different Privacy Claims



WINDOW CLAIM (DIRECT)

(0.53683, 10⁻⁶)

Requested unit matches the accounted unit.

RAW-EVENT CLAIM (CONVERT)

K = 4 → (2.14731, 1.0642 × 10⁻⁵)

A raw-event replacement changes up to K = 4 generated-window records.

PATIENT / OWNER CLAIM (BLOCK)

K = 600 public cap (565 observed) → converted delta ≥ 1

The evidence cannot support a meaningful positive owner claim.

Source: post-execution privacy-claim validation study, WISDM case and Table 3.



Real-World Mapping Evidence: Sepsis and Backblaze

Sepsis establishes medical relevance, Backblaze tests scale and transformation semantics.

SEPSIS · PHYSIONET/CINC 2019

Original use: early sepsis prediction from hourly ICU measurements

Illustrative patient-file schema — not an actual patient record

Hour	HR	O2Sat	Temp	SBP	Lactate	Label
1	86	98	—	120	—	0
2	91	96	37.2	110	2.1	0
3	103	94	38.1	96	3.4	1

patient → hourly event → overlapping 6 h windows

6 h dense: 167,206 windows · $k_e=6$ · $K=12$ · $k_{owner}=331$

Role: medical privacy-unit relevance

BACKBLAZE · DRIVE STATS Q1 2025

Original use: HDD/SSD reliability analysis from daily SMART snapshots

Illustrative drive-day schema — not an actual drive record

Date	Serial	Model	Fail	SMART 5	SMART 194
01-01	D001	Model-A	0	0	31
01-02	D001	Model-A	0	0	32
01-03	D001	Model-A	0	1	34

drive → daily SMART event → ordered/calendar windows

7 d dense: 25.9M windows · $k_e=7$ · $K=14$ · $k_{owner}=84$

Ordered vs calendar: 36,421-window difference

TABLE · MAPPING-ONLY DIAGNOSTICS — NO ϵ/δ CERTIFICATE OR DIRECT VERDICT

Dataset	Mapping policy	Generated records	obs. κ_{event}	candidate event K	obs. κ_{owner}
Sepsis	6h dense	167,206	6	12	331
Sepsis	6h non-overlap	29,958	1	2	56
Sepsis	public owner cap 1	5,000	1	2	1
Backblaze	7d dense, ordered snapshots	25,904,370	7	14	84
Backblaze	7d dense, calendar-contiguous	25,867,949	7	14	84
Backblaze	7d non-overlap, ordered	3,709,484	1	2	12
Backblaze	7d non-overlap, calendar	3,707,669	1	2	12
Backblaze	60d non-overlap, ordered	304,906	1	2	1
Backblaze	60d non-overlap, calendar	302,760	1	2	1

Sources: PhysioNet/CinC 2019, Backblaze Drive Stats, and post-execution study Table 4. Example rows are illustrative.



Evidence-Bound Claim Validation



AGGREGATE RESULTS OVER THE 200 STRUCTURED CASES · TABLE 2

Probe / full contract	Unsafe-positive frac.	Valid-exact frac.	Exact-tuple frac.	Principal evidence not checked
B0 Field presence	1.00	0.60	0.00	types, semantics, files, execution, accounting
B1 JSON Schema	0.96	0.60	0.00	cross-field and execution semantics
B2 Multiplicity only	1.00	0.80	0.00	stability/adjacency, staleness, mechanism
B3 Hash only	0.96	0.60	0.00	consistently wrong semantics or mechanism
B4 Accountant only	0.88	0.60	0.00	mapping, claimed unit, dependencies, runtime
A1 Runtime assumed matched	0.12	1.00	0.94	independent runtime/artifact equality
A2 FULL minus vacuity	0.04	1.00	0.98	reportability of converted numerics
FULL	0.00	1.00	1.00	registered-contract assurance boundary

WHAT B0-A2 MEAN

B0 fields · B1 schema
B2 multiplicity · B3 hash · B4 accountant
→ partial evidence projections

A1: FULL, but runtime match is assumed
A2: FULL, but the δ -vacuity gate is removed
FULL: every obligation is checked together

TEST SUITE CONSTRUCTION · 200 CLAIM-EVIDENCE CASES

VALID 100 = 5 claim paths × 5 scale levels × 4 mapping contexts

Paths: Direct-window · Direct-event · Direct-owner · Convert-event · Group-owner
Mappings: canonical · irregular supports · non-selected decoy rows · reordered files

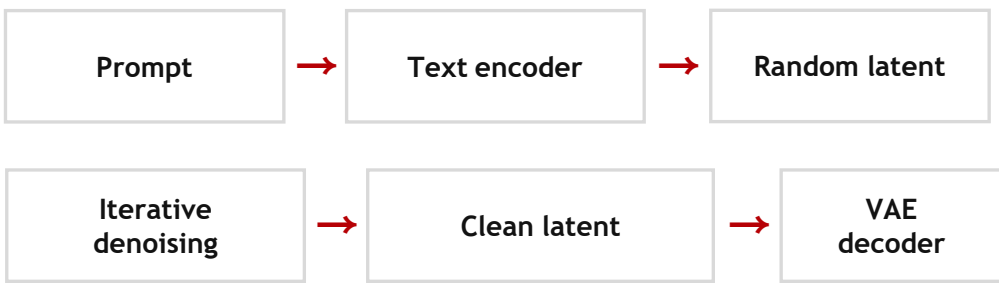
INVALID 100 = 25 semantic mutations × 4 valid parent cases

Partial checks allow 88-100% unsafe positives,
FULL yields 0/100 unsafe, 100/100 valid, and 200/200 exact tuples.

Why AI Contents Need Durable Provenance

Realistic generation creates a provenance obligation, but policy labels still require durable, attack-resistant evidence.

MINIMAL LATENT-DIFFUSION VIEW



Text controls what is generated, the initial latent and denoising trajectory determine the resulting artifact.

POLICY AND PROVENANCE INFRASTRUCTURE



EU AI Act · Article 50
Machine-readable marking



C2PA 2.4
Signed provenance manifests



AT GENERATION

prompt · seed · model
producer · parameters

AFTER DISTRIBUTION

metadata may be stripped
pixels may be recompressed or cropped

AT PUBLIC VERIFICATION

claim may be detached or forged
verifier itself may be exposed

Trustworthy provenance requires evidence that survives distribution and resists verifier aware attacks.



A Public Verifier Becomes an Attack Surface

Open verification removes dependence on a central API, but exposes the decision rule to repeated observation and optimization.

1 · PUBLISH

detector / decoder / queries

2 · OBSERVE

gradients

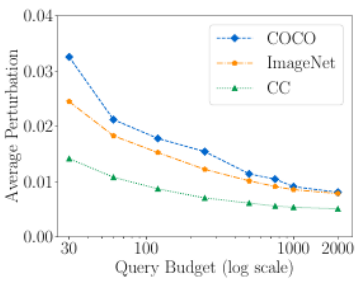
3 · OPTIMIZE

latent watermark

4 · FORGE OR ERASE

pixel watermark

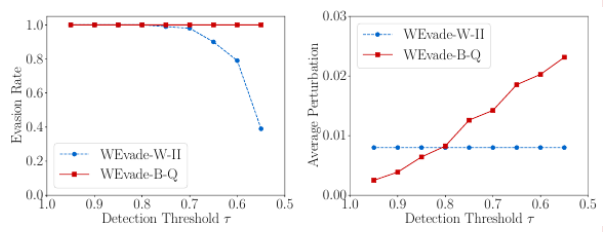
Jiang et al. · CCS 2023



BLACK-BOX QUERY

- ACCESS** binary/score API only (repeated query attack)
- METHOD** repeat queries and update the image
- GOAL** cross the detector threshold

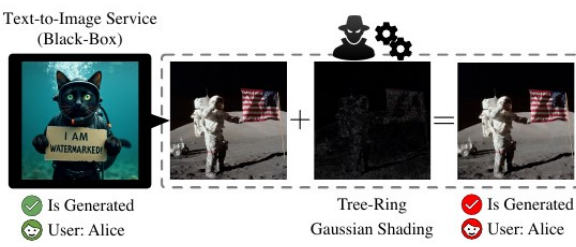
Jiang et al. · CCS 2023



WHITE-BOX GRADIENT

- ACCESS** detector and gradients (direct optimization attack)
- METHOD** use the score as a PGD-style loss
- GOAL** forge or erase within a small budget

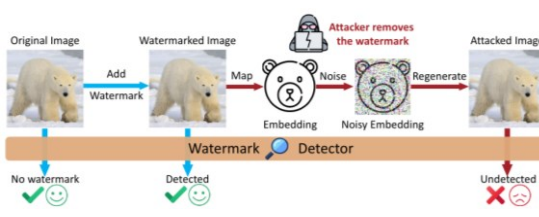
Müller et al. · CVPR 2025



IMPRINT / REPLAY

- ACCESS** one valid watermarked Reference (watermark forgery)
- METHOD** transfer its latent imprint to a cover
- GOAL** false attribution to the reference source

Zhao et al. · NeurIPS 2024



REMOVAL / REGENERATION

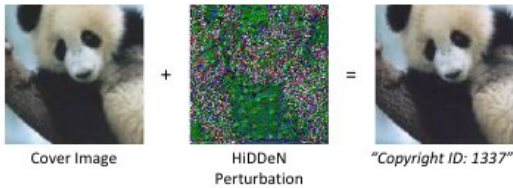
- ACCESS** a watermarked distributed copy (watermark removal)
- METHOD** noise the representation and regenerate
- GOAL** preserve content while suppressing signal



Where Existing Provenance Approaches Break

POST-HOC WATERMARK

HiDeN · ECCV 2018
StegaStamp · CVPR 2020
TrustMark · ICCV 2025
InvisMark · WACV 2025



STRENGTH

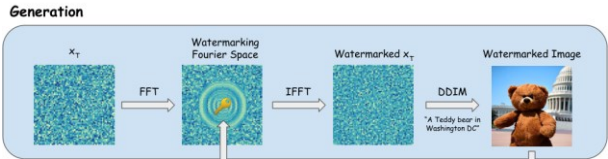
Adds an identifier to any completed image without modifying the generator.

LIMIT

Pixel signals can be removed, regenerated, or optimized against once detection is exposed.

GENERATION-AWARE MARK

Tree-Ring · NeurIPS 2023
Gaussian Shading · CVPR 2024
RingID · ECCV 2024
WIND · ICLR 2025



STRENGTH

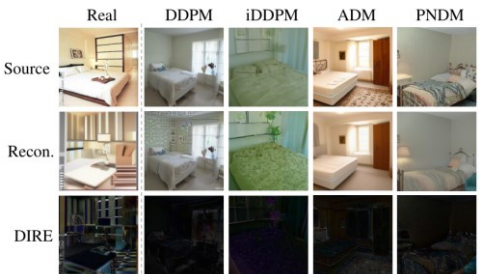
Embeds the signal in the initial latent or generation process, improving binding and robustness.

LIMIT

A public score can still be optimized or transferred and cannot alone prove exact origin.

AI DETECTOR

DIRE · ICCV 2023
UniversalFakeDetect · CVPR 2023



STRENGTH

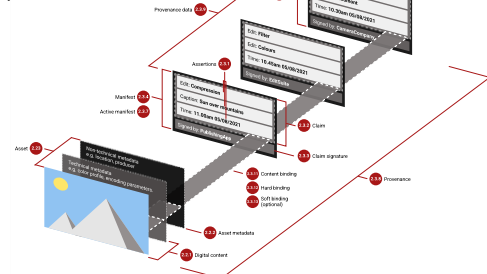
Screens existing images without producer cooperation or a pre-embedded signal.

LIMIT

Unseen generators or adversarial edits can shift learned cues; specific origin remains unproven.

SIGNED METADATA / C2PA

C2PA Content Credentials
Technical Specification 2.4 · 2026



STRENGTH

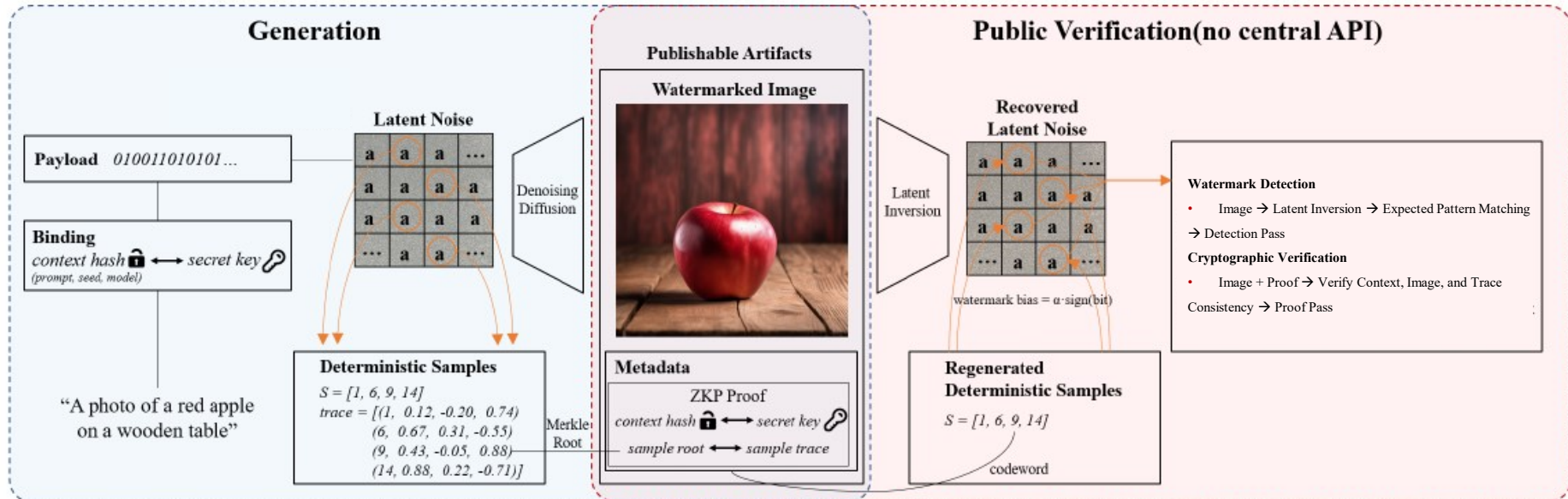
Carries signed history and signer identity, allowing users to trace the declared origin and editing process.

LIMIT

Signed metadata does not prove that embedding occurred.

Needed together: durable in-image signal + public evidence + an acceptance rule that score optimization cannot satisfy.

PP-Mark: Bind Context to the Embedding Trace



1 · CONTEXT BINDING

Prompt · seed · model · producer key
derive binding b and the payload.

2 · EMBEDDING + COMMITMENT

The pattern is embedded in the initial latent;
sampled trace values are Merkle-committed.

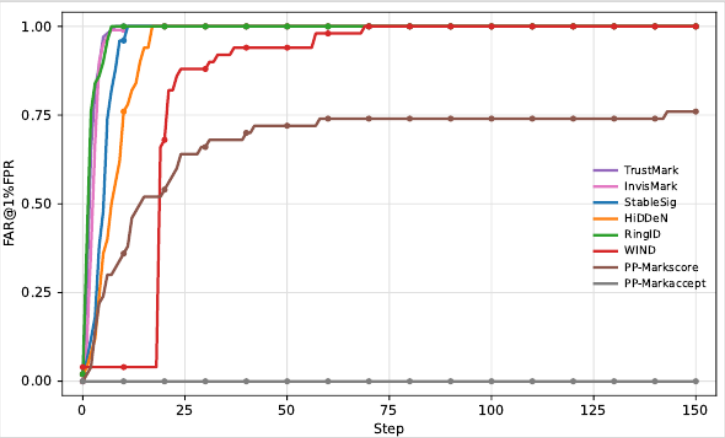
3 · PUBLIC VERIFICATION

Score tests the durable signal; Accept also
requires an image/context/trace-bound receipt.



Main Security Results vs Baselines

WHITE-BOX ATTACK

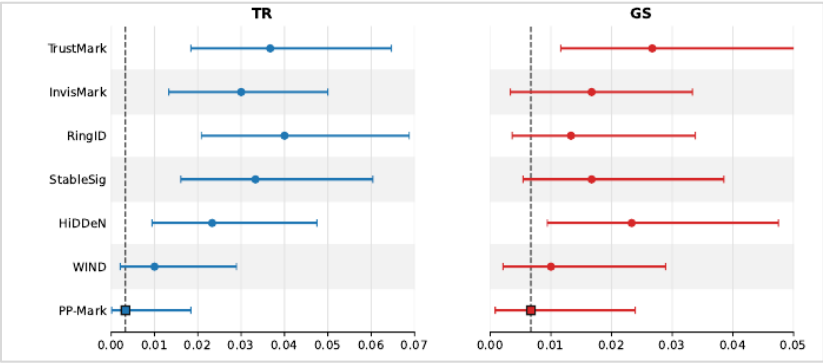


Cumulative FAR@1%FPR vs. step

Method	FAR@1%FPR	Avg. Steps to FA	Avg. PSNR ↑
TrustMark	1.00	2.41	75.73
InvisMark	1.00	3.07	72.67
StableSig	1.00	5.40	65.91
HiDeN	1.00	8.08	62.96
RingID	1.00	2.26	74.51
WIND	1.00	22.20	55.76
PP-Mark _{score}	0.76	17.42	67.69
PP-Mark _{accept}	0.00	—	—

White-Box forgery outcomes across methods. Avg Steps to FA report s, among forgery acceptances, the mean optimization step at which score >threshold first occurs. PSNR is averaged against the original c over image

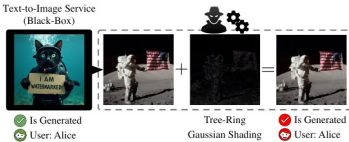
IMPRINT-FORGERY ATTACK



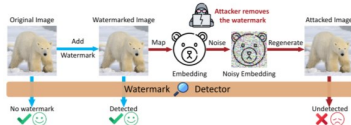
Imprint forgery attack cross method at FPR=1%, Error bars denote 95% CI, the vertical line indicates PP-Mark

Step	TrustMark	InvisMark	RingID	StableSig	HiDeN	WIND	PP-Mark
50	0.05	0.06	0.01	0.04	0.04	0.01	0.00
100	0.03	0.01	0.10	0.04	0.02	0.01	0.00
150	0.03	0.02	0.01	0.02	0.01	0.01	0.01
50	0.01	0.03	0.01	0.01	0.01	0.01	0.00
100	0.02	0.01	0.02	0.02	0.03	0.01	0.01
150	0.05	0.01	0.01	0.02	0.03	0.01	0.01

Imprint forgery attack summary over step. FAR@1%FPR, all methods evaluated at FPR=1%(clean calibrated)



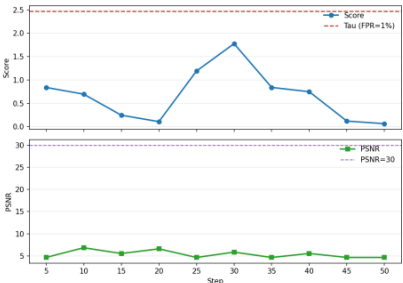
POST-REGENERATION ATTACK



METHOD	NOISE STEP		
	30STEPS	60STEPS	100STEPS
STABLESIG	0.04	0.02	0.03
HiDeN	0.10	0.06	0.04
TRUSTMARK	0.85	0.63	0.39
INVISMARK	0.01	0.05	0.02
TREE-RING	0.95	0.93	0.93
PP-MARK	0.98	0.98	0.98

Post-Regeneration attack detection rate at FPR=1% thresholds, each noise step uses 100 images, noise step is 30/60/10

BLACK-BOX ATTACK



Black-box (SPSA) removal trace. Top: score vs step with τ (FPR=1%); bottom: PSNR vs step.

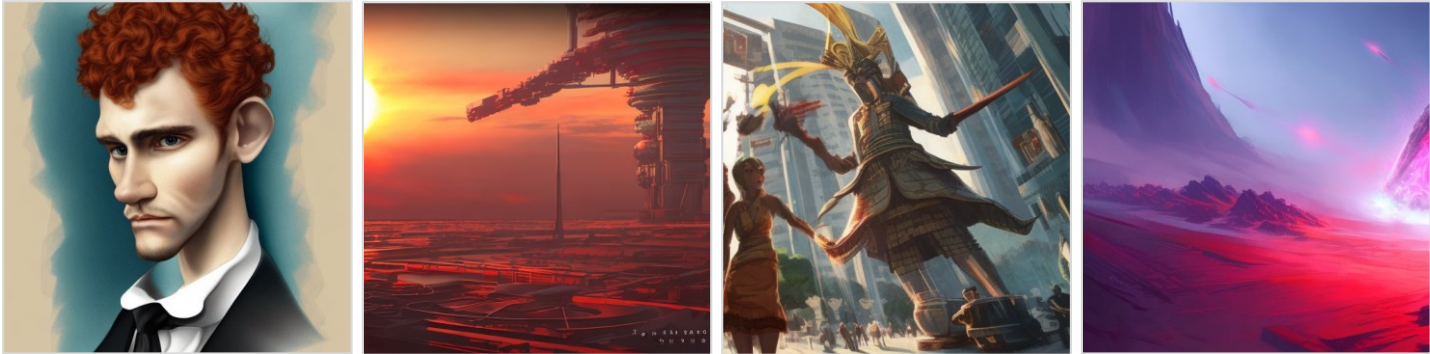
Step	FAR@1%FPR	PSNR ↑
50	0.02	6.55
100	0.00	6.61
150	0.01	6.63
Avg.	0.01	6.60

SPSA-based forgery attack on PP-Mark.



Practicality Experiments

Quality, Costs



Watermarked Image Examples

Method	B95 ↓	KIDx10^3 (->0)	CLIPx100 ↑	Embed(s)	Verify(s)
PP-Mark	34.14	1.101	31.64	0.89	0.75
Gaussian Shading	27.35	0.621	32.42	3.10	0.72
HiDDeN	105.86	7.239	31.30	0.001	0.001
InvisMark	31.36	-0.900	32.53	0.002	0.009
RingID	34.64	-0.989	32.41	0.89	0.72
Stable Signature	32.69	-0.814	32.50	0.91	0.008
Tree-Ring	41.90	-0.090	32.30	0.89	0.75
WIND	27.42	-0.386	32.28	0.89	1.10
TrustMark	29.44	-0.174	32.54	0.013	0.004

Proof configuration	Genaration Time	Artifact
SP1 v6.3.1	17.0 s	2.81 MB
RISCO	180.9 s	40.51 MB
PLONK	312.677 s	3,874 B
Groth16	438.380 s	1,499 B

Robustness



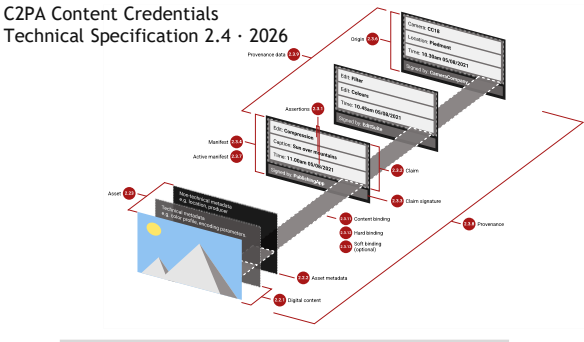
(a) 75° Rotation (b) 25% JPEG Compression (c) 75% Random Cropping + Scaling (d) Gaussian Blurring with an 8 × 8 Filter (e) Gaussian Noise with $\sigma = 0.1$ (f) Color Jitter with a Brightness Factor of 6

Transform	SD 2.1	SDXL
JPEG compression ($q=25$)	0.94	1.00
Gaussian blur (8×8)	1.00	1.00
Additive noise ($\sigma=0.1$)	0.86	0.95
Brightness jitter ($\times 0.6$)	0.98	1.00
Random crop + resize (75%)	0.87	0.74
Rotation 75°	0.85	0.97



VS. C2PA Manifest Signature

SIGNED METADATA / C2PA



STRENGTH

Carries signed history and signer identity, allowing users to trace the declared origin and editing process.

LIMIT

Signed metadata does not prove that embedding occurred.

Test	Bare-Sig	Context-Sig	PP-Mark (detection)
Signed unwatermarked image	100% Accepted	100% Accepted	0% Accepted
JPEG q25	0% Valid	0% Valid	94% Valid
75% crop-resize	0% Valid	0% Valid	87% Valid

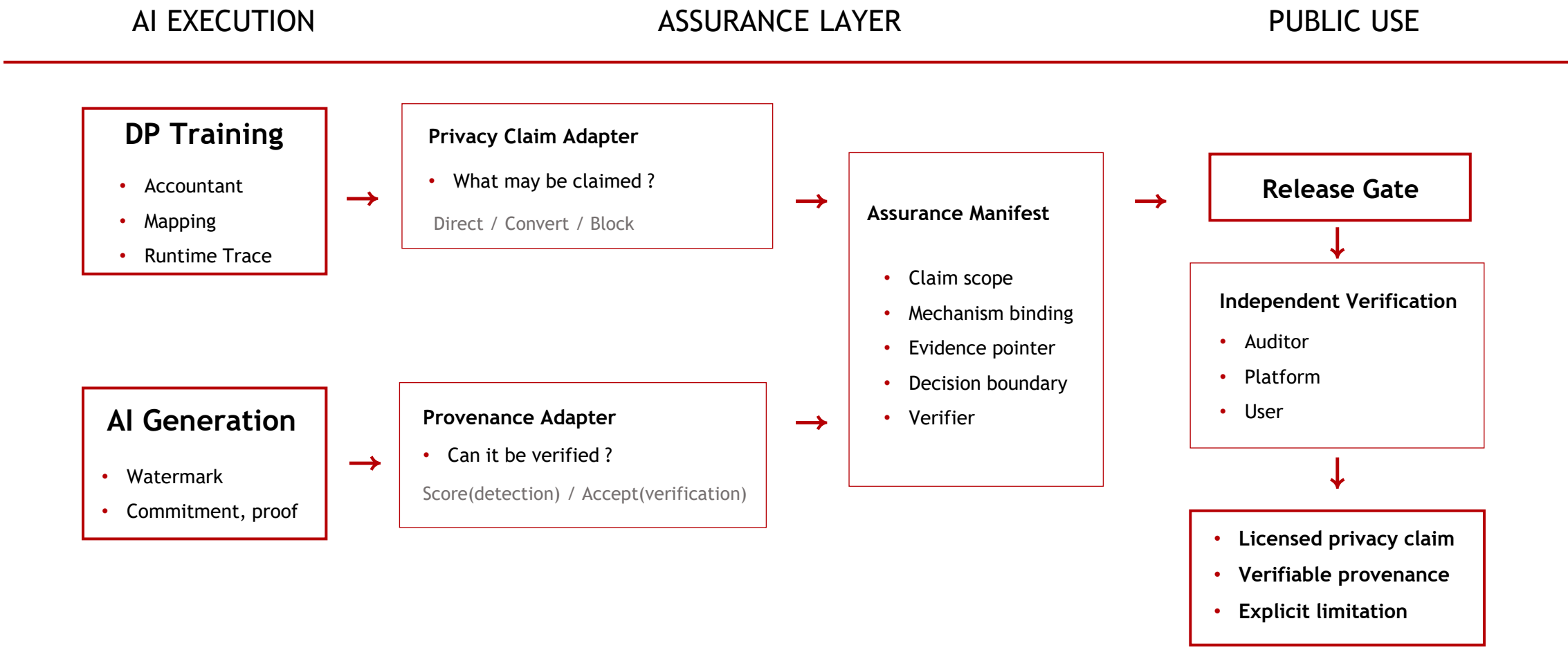
Attack	Sig-only	Sig+Score	PP-Mark (detection)	PP-Mark (verification)
Forged binding	100%	100%	0%*	0%
No embedding	100%	0%	0%*	0%

* 0/100 observed, nominal FPR <= 1%

Threat model	PP-Mark (detection)	PP-Mark (verification)
External adversary (unknown k)	✓	✓
Generator fraud (no embedding)	✗	✓
Third-party audit (no k disclosure)	✗	✓



Integrated Blueprint: AI Artifact Release and Verification Platform





Future Studies: Closing the Remaining Assurance Gaps

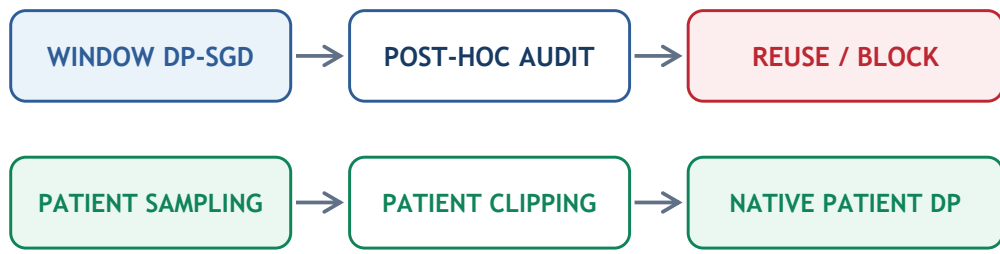
WHEN IS NATIVE PATIENT DP WORTH THE COST ?

Reuse an existing window-level result, or retrain for a direct patient guarantee?

5,000 PhysioNet Sepsis patients / 370 positive patients



SAME PATIENT SPLIT, 6-HOUR WINDOWS



converted privacy / membership inference / time and memory

RECOVERING PP-MARK PROOFS AFTER DISTRIBUTION

Store the proof at publication, retrieve the correct proof when verification is requested.

AT GENERATION



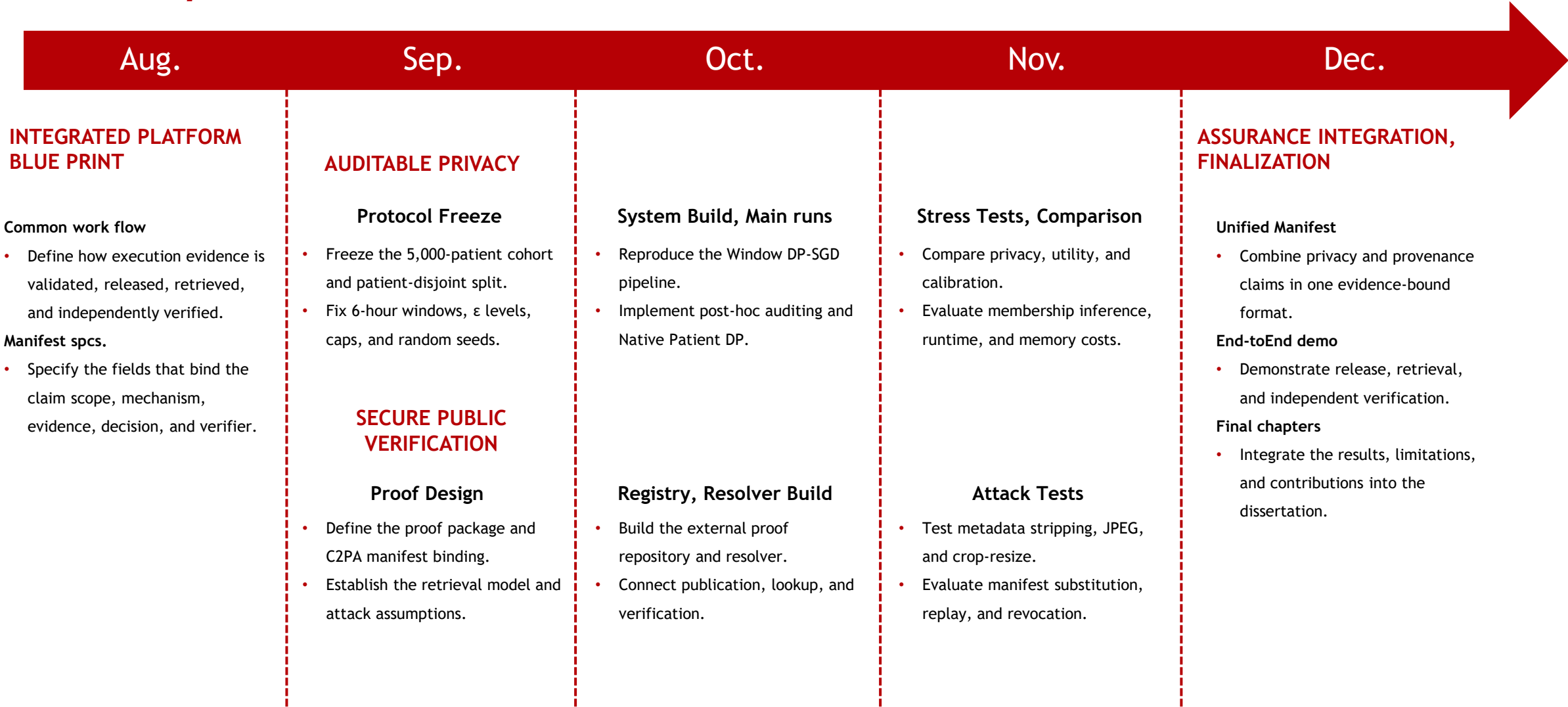
LATER VERIFICATION



TEST: data stripping / JPEG / crop-resize / manifest swap / proof replay



Next Steps: Research Timeline



Thank You.